SALES DATA ANALYSIS USING SQL

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BATCH -DA/DS 39

INTRODUCTION

- Analysis of sales dataset using SQL
- Data includes customer info, category, quantity, and sales amount
- Three stages:
- Data Cleaning
- Data Exploration
- Data Analysis
- Goal: Generate business insights for better decision-making

OBJECTIVE

1. Clean sales dataset by removing nulls/incomplete data.
2. Explore key metrics (sales count, customer count, categories).
3. Answer business questions using SQL queries.
4. Identify best-selling categories, top customers, and trends.

TOOLS & TECHNOLOGIES

- Database: MySQL WORKBENCH
- SQL Queries for:
- Data Cleaning
- Data Exploration
- Data Analysis

DATASET OVERVIEW

- **Table: salestable**
- **Columns: sale_date, sale_time, customer_id, gender, age, category, quantity, total_sale**
- **Brief about data volume: 1,987 sales, 155 unique customers, 3 categories (Clothing, Beauty, Electronics).**

DATA LINK-https://github.com/najirh/Retail-Sales---P1/blob/main/SQL%20-%20Retail%20Sales%20Analysis_utf%20.csv

DATA CLEANING

- Removed rows with NULL values in key fields
- Used DELETE statements for incomplete data
- Enabled sql_safe_updates = 0 for safe deletion

```sql
select*from salestable;
use project5;
select * from salestable where sale_date is null;
select * from salestable where sale_date or age or gender or customer_id is null;
delete from salestable where sale_date is null
or customer_id is null
or sale_time is null
or gender is null
or age is null
or category is  null
or quantiy is null;
set sql_safe_updates=0;
```

DATA EXPLORATION

- Total sales: 1,987
- Unique customers: 155
- Categories found: Clothing, Beauty, Electronics
- SQL used: COUNT, DISTINCT

```
##dataexploration
--how many sales we have;
select count(*) as total_sale from salestable;
--we have 1987 sales;
-- how many customers we have;
select count(*) as customer_id from salestable;
select count(distinct customer_id) from salestable;
```

```
select distinct category from salestable;
categories are-
clothing
beauty
electronics;
```

Result Grid

total_sale
1987

Result Grid | Filte

count(distinct customer_id)
155

Result Grid | Filter Rc

category
Clothing
Beauty
Electronics

BUSINESS QUERIES (PART 1)

- Sales on specific date (05-11-2022)
- Clothing category sales > 4 in Nov 2022
- Total sales for each category

```
Q1- write sql query to retrieve all columns from sales made on 05-11-2022;
select*from salestable where sale_date='05-11-2022';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ï»¿transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantiy	price_per_unit	cogs	total_sale
180	05-11-2022	10:47:00	117	Male	41	Clothing	3	300	129	900
240	05-11-2022	11:49:00	95	Female	23	Beauty	1	300	123	300
1256	05-11-2022	09:58:00	29	Male	23	Clothing	2	500	190	1000
1587	05-11-2022	20:06:00	140	Female	40	Beauty	4	300	105	1200
1819	05-11-2022	20:44:00	83	Female	35	Beauty	2	50	14	100
943	05-11-2022	19:29:00	90	Female	57	Clothing	4	300	318	1200

```
select*from salestable where category='clothing' and sale_date=11-2022 and quantiy <=4;

there is no more than 4;
```

```
select category, SUM(total_sale) as net_sales from salestable group by category;
```

Result Grid | Filter Rows:

category	net_sales
Clothing	309995
Beauty	286790
Electronics	311445

BUSINESS QUERIES (PART 2)

4. Average age for Beauty purchases
5. Transactions with sales > 1000
6. Transactions count by gender & category

```
Q4- write a sql query to find the average age of customers who purchased beauty category;

select avg (age) from salestable where category='beauty';
```

Filter Rows:

avg_sale
457.0861

```
q6) write a sql query to find the total number of transactions made by each gender in each category;

select gender,category ,count(*) from salestable group by gender,category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantiy	price_per_unit	cogs	total_sale
522	09-07-2022	11:00:00	52	Male	46	Beauty	3	500	145	1500
559	12-12-2022	10:48:00	5	Female	40	Clothing	4	300	84	1200
1522	14-11-2022	08:35:00	48	Male	46	Beauty	3	500	235	1500
1559	20-08-2022	07:40:00	49	Female	40	Clothing	4	300	144	1200
421	08-04-2022	08:43:00	66	Female	37	Clothing	3	500	235	1500
1421	17-01-2022	07:07:00	59	Female	37	Clothing	3	500	185	1500


```
q6) write a sql query to find the total number of transactions made by each gender in each category;

select gender,category ,count(*) from salestable group by gender,category;
```


Result Grid | Filter Rows:

gender	category	count(*)
Male	Clothing	351
Male	Beauty	281
Female	Clothing	347
Male	Electronics	343
Female	Beauty	330
Female	Electronics	335

Result 11

BUSINESS QUERIES (PART 3)

7. Average sales per month & best-selling month/year
8. Top 5 customers by total sales

```
Q7)write a sql query to calculate the average sale for each month.find out the best sale month in each year;

select
extract(year from sale_date) as year,
extract(month from sale_date) as month,
avg(total_sale)as avg_sale
from salestable
group by 1,2
order by 1,3 desc;
```

```
Q8)write a sql query to find the top 5 customers based on the highest total sales;

select customer_id,SUM(total_sale) as total_sales
from salestable group by 1
order by 2 desc
limit 5;
```

RESULTS & INSIGHTS)

- Clothing category has highest/lowest sales (based on data)
- Best-selling month/year identified
- Top 5 high-value customers listed
- Gender & category purchase patterns

CHALLENGES & SOLUTION

- Data with null values → removed
- Incorrect date format → standardized before queries
- Needed to group and aggregate efficiently → used GROUP BY & functions

CONCLUSION AND REFERENCES

- Project demonstrates SQL's capability in cleaning, exploring, and analyzing sales data.
- Provided actionable business insights.

- SQL documentation (MySQL/your DB)
- Dataset source -github
- Online resources used